

1. Input Modalities

Pre-Event SAR

Dual-Pol (VV, VH) dB

Post-Event SAR

Dual-Pol (VV, VH) dB

Topographic Geometry g

$\cos(\text{LIA})$, Slope, DEM

Radar-Aspect Vector

$\cos(\alpha - \phi_{\text{look}})$, $\sin \alpha$, $\cos \alpha$

2. Feature Pyramids

Siamese SAR Encoder

$\mathcal{E}_I$: Shared ResNet Stages
 $C_I \in \{32, 64, 128, 256\}$

Bi-temporal Fusion

$\mathbf{F}_{\text{diff}}^I = [\mathbf{F}_{\text{post}} - \mathbf{F}_{\text{pre}}, \mathbf{F}_{\text{post}}, \mathbf{F}_{\text{pre}}]$

Topo Pyramid

$\mathbf{T}^I$: Multi-scale Conv
Matches $C_I \times H_I \times W_I$

3. RTCAB Attention

SAR Queries Q

$\mathbf{F}_{\text{diff}}^I \mathbf{W}_Q$ (Dense Tokens)

Topo Keys / Values

$\text{Pool}_{8 \times 8}(\mathbf{T}^I) \mathbf{W}_{K, V}$

Cross-Attention Gate

$\mathbf{A} = \text{softmax}(\mathbf{Q}\mathbf{K}^T / \sqrt{d_k}) \mathbf{V}$

Spatial Topo Gate

$\mathbf{W}_{\text{gate}} = \sigma(\text{Conv}(\mathbf{T}^I))$
 $\tilde{\mathbf{F}}^I = (\mathbf{F}_{\text{diff}}^I + \mathbf{A}\mathbf{V}) \odot (1 + \mathbf{W}_{\text{gate}})$

4. Decoder

Progressive Decoder

ConvTranspose2d + Skips
Hierarchical Upsampling

Predicted Mask $\hat{\mathbf{Y}}$

Continuous Probabilities
Full-Scene Blending
1.80s / Scene Latency

5. Physical Loss

Segmentation $\mathcal{L}$

Weighted BCE + Dice
 $w_{\text{pos}} = 3.0$ (Imbalance)

Geo-Loss $\mathcal{L}_{\text{geom}}$

Penalizes mass in:
Valley floor $< 5^\circ$
Sheer cliff $> 65^\circ$

Total Loss $\mathcal{L}$

$\mathcal{L}_{\text{BCE}} + \mathcal{L}_{\text{Dice}} + 0.5\mathcal{L}_{\text{geom}}$